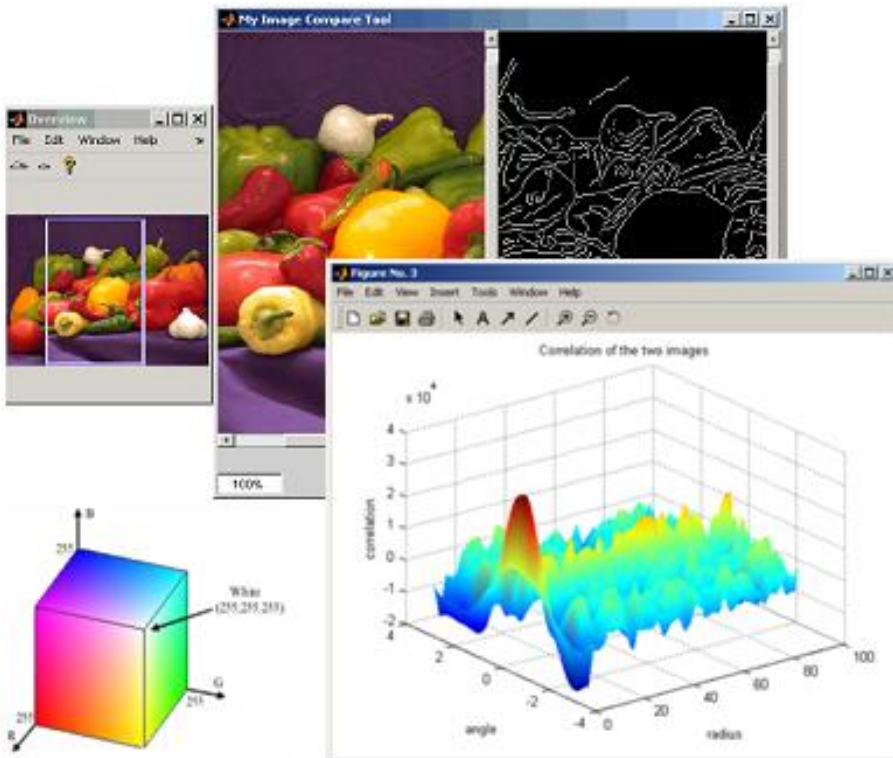


# Digital Image Processing

## SPATIAL FILTERING (UNIVERSITY QUESTIONS)



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# Smoothing Operation

# Numerical Examples

**Ques1:** Consider the following image

|   |   |   |   |   |
|---|---|---|---|---|
| 0 | 1 | 0 | 2 | 7 |
| 2 | 1 | 6 | 1 | 0 |
| 5 | 6 | 7 | 6 | 3 |
| 1 | 1 | 6 | 1 | 5 |
| 5 | 4 | 2 | 2 | 5 |

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What will be the new value to the pixel (2,2) if the smoothing is done using a  $3 \times 3$ :

- a. Mean Filter
- b. Weight average filter (Assign weight as 3, 2 and 1)
- c. Median filter
- d. Min filter
- e. Max filter

# Numerical Examples

**Ques2:** Consider the following image. What will be the new pixel value of the pixel (1,1) if the smoothing is done using a  $3 \times 3$

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|     |     |     |     |
|-----|-----|-----|-----|
| 240 | 111 | 150 | 93  |
| 75  | 199 | 80  | 156 |
| 20  | 42  | 30  | 5   |
| 5   | 77  | 230 | 60  |

- a. Mean Filter
- b. Median Filter
- c. Weight Average Filter

# Numerical Examples

**Ques 3:** Consider the following image. What will be the new value of the pixel (2,2) if smoothing is done using a  $3 \times 3$

|     |     |     |    |
|-----|-----|-----|----|
| 240 | 11  | 100 | 93 |
| 124 | 178 | 229 | 56 |
| 20  | 42  | 190 | 0  |
| 5   | 77  | 230 | 60 |

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- a. Mean filter
- b. Min Filter
- c. Median Filter

# Numerical Examples

**Ques 4:** if  $X = \{2 \ 3 \ 4 \ 3 \ 4 \ 5 \ 6\}$ , and  $W = \{1 \ \underline{1} \ 1\}$ . What would be the result of median filtering? Use zero padding if required.

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# Sharpening Operation

# Numerical Examples

**Ques1:** Compute the convolution of the Laplacian Kernel  $L_4$  with the image given below. Use Border values to extend the image.

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|    |    |    |    |    |
|----|----|----|----|----|
| 50 | 50 | 50 | 50 | 50 |
| 50 | 50 | 50 | 50 | 50 |
| 10 | 10 | 10 | 10 | 10 |
| 10 | 10 | 10 | 10 | 10 |
| 10 | 10 | 10 | 10 | 10 |



# Numerical Examples

**Ques2:** Compute the convolution of the Laplacian Kernel L\_8 with the image given below. Use border values to extend the image.

**GLA University, Mid term Examination, Even Semester 2022-23**

|     |     |     |     |
|-----|-----|-----|-----|
| 100 | 100 | 150 | 150 |
| 100 | 100 | 150 | 150 |
| 100 | 100 | 150 | 150 |
| 100 | 100 | 150 | 150 |
| 100 | 100 | 150 | 150 |

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[illegible]

# Numerical Examples

**Ques4:** Compute the convolution of the Laplacian Kernel L\_8 with the image given below. Use Border values to extend the image.

**GLA University, Mid term Examination, Odd Semester 2023-24**

|     |     |    |    |    |
|-----|-----|----|----|----|
| 100 | 100 | 10 | 10 | 10 |
| 100 | 100 | 10 | 10 | 10 |
| 100 | 100 | 10 | 10 | 10 |
| 100 | 100 | 10 | 10 | 10 |
| 100 | 100 | 10 | 10 | 10 |

# Numerical Examples

**Ques5:** A mask is defined with the following mathematical expression:

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$$f(x-2,y) + 2f(x-1, y) + 2f(x+1, y) + f(x+2, y) + f(x, y-2) + 2f(x, y-1) + 2f(x, y+1) + f(x, y+2) - 12f(x, y).$$

Draw this mask.

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 9 | 9 | 9 | 8 | 7 | 6 | 5 | 4 | 2 | 2 | 2 | 2 | 8 | 8 | 6 | 6 | 6 | 6 | 6 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

Your answer should be in tabular form having 3 rows as shown below. First row should show original signal, second row should have values for the first-order derivative and the third row should have values for the second order derivative

[illegible]

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 6 | 6 | 6 | 6 | 5 | 4 | 3 | 2 | 1 | 1 | 1 | 1 | 1 | 1 | 6 | 6 | 6 | 6 | 6 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

Your answer should in the tabular format having 2 rows as shown below. First row should be the original signal. Second rows should have values of the First Order Derivative.

[illegible]



Thank You